Measuring all the noises of LLM Evals

[Draft Dec 20th – feedback welcome]
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Abstract

As LLM benchmark questions grow more complex, evaluation sample sizes have decreased, heightening the risk of being fooled by randomness. Well-established statistical methods are often misapplied or omitted in LLM research. In this work, we apply the more powerful paired methods to many models and agents evaluated on many benchmarks, making these statistical methods useful to practitioners without much additional effort. Conceptually, we adapt them to LLM evaluations by clearly defining and measuring the prediction noise, the data noise, and their combined total noise. The resulting sample estimators are tested extensively on generate models and are shown to be consistent with the bootstrap and sign test. By analyzing millions of question-level results across hundreds of LLMs and agents on popular benchmarks, we find that each benchmark exhibits a characteristic and highly predictable total noise level on all model pairs. This finding allows practitioners to benefit from rigorous statistical analysis without having to do it themselves and even to infer the likely significance of the results reported by others.

1 Introduction

Inferring the signal from noisy measurements is a key of experimental science. The AI effort is now a leading AI benchmarks have grown much harder yet are much smaller from 10000 examples on MNIST to 500 on SWEBench-Verified, which heightens the risk of being fooled by randomness. Well-established methods from statistics *can increase the rigor of the conclusions drawn from data* (Benjamini et al., 2021). Even though these methods are found in landmark textbooks and is described clearly for LLM evaluations (Miller, 2024), they are not used by important papers, used incorrectly, or unnecessarily loose (Example 2). While these carefully considered and well-studied methods have long been understood by some experts, they are not understood widely enough to be used correctly and commonly. We apply these well-studied methods to all models and agents evaluated by many benchmarks, and output the resulting noise measurements. Due to the predictability in these measurements, practitioners can use well-tested and accurate-enough noise measurements without running a program or understanding the details of the methods. Besides better usability, this approach also allow us to infer the likely significance of results reported by others.

To do this, we define and measure the *prediction noise* when answering a given question differently in different prediction samples, the *data noise* due to using a finite sample from the population of possible questions, and the *total noise* following the law of total variance. The prediction noise can be observed directly and is significant in LLMs at a higher temperature. The data noise is often ignored in LLM development (Example 1), perhaps because it cannot be measured directly by fixed evaluations. For most cases we argue for the use of total noise as the best standard.

As more resources are devoted to model capability, it becomes more expensive to reject significant results due to a weak analysis. While improving models is hard, the more powerful paired analysis is a known method that should be used. The paired methods considers how model predictions are different from each other. A difficulty is they only apply to pairs of results rather than directly to a table or leaderboard. Our all-pairs-paired method applies the paired method to all pairs of LLMs anyway –

Preprint. Under review.

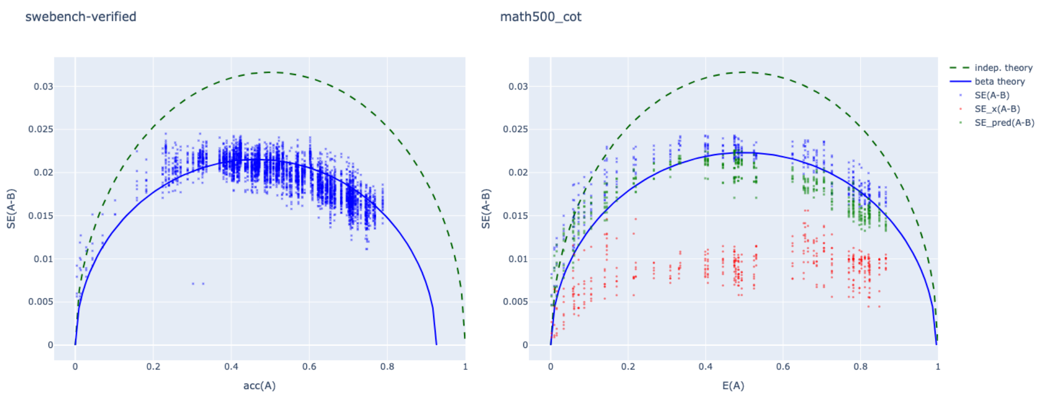

Figure 1: paired total standard errors vs. the accuracy, showing a clear trend in the empirical results agreeing with the beta theory prediction. Left: on SWEbench-verified 1 prediction per example so only the total noise is estimated, Right: MATH500 with 1000 predictions per question and estimated data noise SE_x, the prediction noise SE_pred, and total noise SE. More details in Section 4.

we compute the paired variance $\mathrm{Var}[A-B]$ for all model pairs A, B based on the question-level predictions on each benchmark. We find that each benchmark has a characteristic noise level for all pairs of LLM or agents, so without doing separate statistical testing for each new experiment, we can predict the paired variance of the benchmark used. In particular, across benchmarks and LLM/agents, the more powerful paired test usually gives similar total variance as the basic variance predicted by the accuracy p_A of model A, giving this rule of thumb on correctness evaluations

$$\mathrm{Var}[A-B] \approx \mathrm{Var}[A] = p_A(1-p_A),$$
$$\mathrm{SE}[A-B] \approx \mathrm{SE}[A] = N^{-1/2}\mathrm{Var}[A]^{1/2}.$$

This allows us to meaningfully use a characteristic noise levels per benchmark, thus we have a good idea of the noise levels without running this analysis for each result reported by others or from our own experiments. In addition, Figure 1 shows an example of reference measurements of all noise types that are carefully validated for correctness. With a decoupling of the noise analysis from particular experiments, it is easier to obtain correct variance values and confidence intervals for LLM evaluations, instead of relying on the experimenter to provide their own analysis for each pair of result, which is often done incorrectly in a hurry as afterthoughts to their main results.

The basic analysis method is presented from first principles (Section **??**), validated by comparing with well-established bootstrap and sign-test (Section 3.5), the implementation (Section 3.4) is tested against several generative models of the ground truth as well as on many common LLM benchmarks and model combinations, where millions of question level results are aggregated. Section **??** describes data visualization methods that were useful for gaining a qualitative understanding of the model predictions. We show that the noise level is predictable on all the benchmarks, under different temperatures and from millions of question level responses based on public leaderboards as well as on controlled experiments. In discussions (Section 6), we describe a simple generative model that explains the observed data based on the $\mathrm{Beta}(1-p_A, p_A)$ distribution, why difficult questions cannot yet make up for decreased sample size, how to combine multiple evaluations based on noise, illustrative examples about noise, and some suggestions for current and future benchmarks.

[TODO: contributions, website reference.]

2 Main concepts and examples

2.1 Type of noise

We should clearly distinguish three types of noise: the prediction sampling noise when answering a given question, the data sampling noise due to using a finite sample from the population of possible questions, and their sum total noise.

Prediction noise. The prediction noise comes from generating different answers on a given question with the same model. LLMs have the remarkable ability to sample independent predictions that are especially diverse on reasoning and agentic questions. In the physical world, the prediction noise typically cannot be measured directly — humans cannot completely forget their previous thoughts and medical treatments cannot be undone to get more independent samples. Previous probabilistic ML models such as classifiers can typically make their best prediction directly and do not have interesting prediction noise.

The prediction noise is also persistent even though we can try to reduce it by fixing the random seed, decreasing the temperature, averaging over many samples per question, or applying an aggregation method such as majority voting and verification. Fixing random seed or low temperature is brittle, where it only sometimes works on the exact same model. Trying to produce the best answer by aggregation is perhaps the most sound method, though this still leave some prediction noise due to the diversity of samples. Among these noise reductions, only averaging and fixing the random seed gives the same mean while others introduce a bias. Averaging is the most general and effective, which we will focus on.

On top of sampling, additional prediction noise can come from inference-time choices such as prompt formatting, hyperparameters, and answer extraction; and training-time choices such as the random seeds, hyperparameters, and training steps. All of which can theoretically be measured by repeating the process of changing the choice, optionally training the model, and then drawing predictions on a given question.

Data noise. The eval data sampling noise comes from sampling a particular N questions instead of another equally good N questions from the unobservable population of questions. As long as the model has different expected accuracy on different questions (e.g. some questions are harder), the average varies with the sampled questions. Unlike the prediction noise, the data noise cannot be directly measured by evaluating on the fixed, given set of questions. Though the data noise can be inferred using the bootstrap or the variance estimator, it is not observed in experiments such as the training curve (Example 1) and varying the training seeds (Madaan et al., 2024).

To see the presence of data noise, suppose that out of 100 True and False questions, model A is correct on 50 random questions, B is correct on 52 random questions out of the same 100 questions. Then B is not reliably better than A even when each question is answered deterministically with no prediction noise. Suppose we have $p = 0.5$ chance of sampling a question that is answered correctly by A, then the number of correctly answered questions have a standard deviation of $(100p(1-p))^{1/2} = 5$, and thus it is typical to see a difference of more than 2 questions. Statistical methods such as computing the standard error or bootstrap focus on the data noise and will show that A is not reliably better in this case.

If we only care about a specific set of questions, then we can ignore the data noise. There are real life examples when the goal is to memorize a specific list of questions such as a driving knowledge test where the set of questions is published, or to solve specific important problems (AGI/ASI, P vs. NP). On most evals, we care about the underlying ability measured by the eval rather than the specific questions of the eval, so the data noise is not to be ignored.

Total noise = prediction + data noise. We can make this relation precise using the law of total variance. The extensive literature about noise focus on the total noise or the data noise, perhaps because the prediction noise is uninteresting as in classification, or not measurable directly as in most non-digital experiments.

Example 1. a) The training curve is insufficient in theory. The *training curve* plots the evaluation result as a function of training steps and graphically shows the variance of evaluation results as a function of training steps (Figure 2, col 1). While separated training curves is intuitively necessary for reliable results, it is not sufficient. The training curve only shows the prediction noise and completely ignores the data noise since the exact same evaluation questions are used. In theory, the data noise can be much higher than the prediction noise. For example, suppose the training data of models A and B consists a random leak of the 100 eval questions, each included with probability 0.5. Suppose that B is 5% more contaminated than A just due to noise. Then by design, the procedure leading to B is not better than A, however B has a consistently higher training curve if both A and B only correctly answer the eval questions leaked into their training set and wrong on the rest, both eventually reaching a constant peak after seeing all the leaked questions. In this example, any statistical method considering the data noise will tell us that B is not better than A.

b) The training curve is valid when prediction noise is dominant. In practice, if the prediction noise dominates the data noise, then the training curve method is valid. Figure 2 shows increasingly powerful analysis on the training curve.

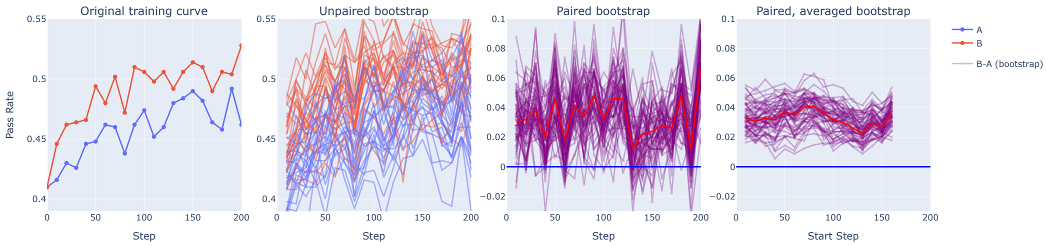

Figure 2: **Column 1)** The original training curve where A and B both starts from the same checkpoint at step 0 where data noise is 0, B appears to be better. **2 unpaired bootstrap)** Many As are higher than many Bs, basically a meaningless difference. **3 paired bootstrap)** Slightly significant with a z-score of 1.7 at 1 prediction per question. **4 paired, averaged bootstrap)** The result is very significant with a z-score of 3.5 if we average the question level predictions over 5 consecutive training checkpoints and reduce the prediction noise.

Paired analysis. Whereas insights, skills and resources are required to improve LLM ability, the paired methods can increase the signal significantly and should be recommended as enormous resources are devoted to improving models. The paired method makes use of prediction differences on the same question to gain more information. To see why paired method is better on real data, Figure 4 shows the heatmap of correctness. We can observe that models at a similar overall accuracy (close in x-axis) also does similarly on the individual questions and thus the paired method has the potential to reduce the data noise significantly. Example 3 shows the intuition on how paired analysis works. Examples 1b and 2 show the potential increase in power using paired methods, especially when prediction noise > data noise, leading to the ability to detect differences several times smaller.

Example 2 (HumanEval)**.** On HumanEval with $N = 164$, if a model has $\bar{A} = 0.5$ accuracy, then the unpaired standard error (SE) is $\sqrt{1/164 \times 0.5(1-0.5)} = 4\%$. Thus we need $4\% \times \sqrt{2} \times 1.96 = 11\%$ to get 0.05 p-value. The table below shows the reductions with paired and averaging. Table 10 of

Test Type	Standard error	Difference for $p < 0.05$
Unpaired (mean = 0.5)	6%	11%
Paired single prediction	4%	8%
Paired with averaging	1–2%	2-4%

Llama 3 (Grattafiori et al., 2024) reports the unpaired 95% confidence intervals of around 7%, but a factor of $\sqrt{2}$ is needed when comparing pairs of models, thus many of the comparisons do not clear the $10\% - 11\%$ difference with this weak analysis method.

Example 3 (Paired vs unpaired)**.** In the unpaired case, suppose A scored 50% on this year's exam, B scored 60% on last year's exam where each exam contains different questions drawn from the same distribution. So B might have done better because last year's exam was easier due to randomness. In the paired case, suppose B scored 60% on the same exact exam as A with exactly the same questions, then the same 10% difference is more indicative of the better performance by B.

All-pairs paired method. We measure the paired standard error $\mathrm{SE}(A-B)$ of all model pairs A and B, where the total noise is always measured, and the data and prediction noises are measured whenever possible. While this approach can be inconclusive in general (Example 4), our measurements show that on LLM evals this turned out to be quite informative, so we highlight the following two findings.

Finding: typically prediction noise > data noise. When this holds, we can gain much statistical power by using the expected metric where the prediction noise is averaged away to be comparable to the data noise remaining. Figure 1 Right show that on MATH500, the data standard error (SE) is typically less than $1/2$ of the prediction SE between different final models. Example 1b and Figure 2 show training curves evaluated on SWEBench-verified starting with the same checkpoint, the data SE is $\sim 1/6$ of the prediction SE. If we average over 36 predictions, then the total SE would be $\sqrt{2}/6 = 0.24$, and allowing us to detect effects that are 0.24 the size with the same p-value and power. Example 2 shows that the smaller reduction on HumanEval is still a factor of 2. For controlled experiments, this allows us to detect differences several times smaller at the same significance level and power. Some caveats are discussed in §6.3.

Finding: predictable total noise. Figure 1 shows the total noise vs. the accuracy of A. We observe that the total noise is quite predictable and agrees well with a theoretical model. This means each eval has a characteristic total noise as a function of the accuracy. This is the right noise number to use for experiments where one prediction is made per question.

Example 4 (Error bars for leaderboards)**.** Gu et al. (2024); Chiang et al. (2024) tried to put error bars on a leaderboard. To do this, one can use the unpaired method, or use a fixed baseline for paired comparison. However, this not generally meaningful: let A be the baseline, B and B' are a pair of similar predictions very different from A, so $\mathrm{Var}[A - B] \gg \mathrm{Var}[B - B']$. This example shows it's not in general possible to have one error bar per model, or per accuracy with the paired method. However, at least for the total noise of LLM evals, this work shows that there is almost no such surprises, thus we can compute meaningful error bars.

3 Method

3.1 Setup and notations

To make these concepts precise, there are N evaluations *questions* $\{x_1, \dots, x_N\}$, which are typically text prompts. The model makes prediction $\hat{y}(x)$ and we get $\mathrm{metric}(\hat{y}, x)$, typically 0 for incorrect and 1 for correct but can be a real number or an aggregate metric. Any information needed to compute the metric such as the ground truth answer or agent environments are included implicitly. We use $A(x) := \mathrm{metric}(\hat{y}(x), x)$ to denote the metric function evaluating model A on the question x.

On a benchmark, we compute the average of all N questions to get the mean metric

$$\bar{A} := \frac{1}{N}\sum_{i=1}^{N} A(x_i) \approx \mathrm{E}_x[A(x)].$$

The estimated variance of A and standard error (SE) of the mean $\bar{A}$ are respectively

$$\mathrm{Var}_x[A(x)] \approx \frac{1}{N}\sum_{i=1}^{N} \left(A(x_i) - \bar{A}\right)^2,$$

$$\mathrm{SE}(A, N) = N^{-1/2} \cdot \mathrm{Var}[A]^{1/2}.$$

SE decreases at a rate of $N^{-1/2}$ and also depends on the population variance $\mathrm{Var}[A]$ where $\mathrm{Var}[A] \leq 1/4$ for binary correctness $A \in \{0, 1\}$. N should be large enough for estimating the variance accurately, so a bias correction on $\mathrm{Var}_x[A]$ using $1/(N-1)$ is not useful.

With models A and B, the score difference $\bar{A} - \bar{B}$ can be compared to $\mathrm{SE}(A - B) = (\mathrm{SE}(A)^2 + \mathrm{SE}(B)^2)^{1/2}$ to determine if the difference is likely due to chance. While this is simple and correct, it is also unnecessarily weak (Example 2).

Paired comparison. Whereas innovation, insight, or compute is required to get real signal, tightening the analysis is easy with the paired methods and should be recommended generally. $A(x)$ and $B(x)$ are correlated when the same question is used to evaluate both $A(x)$ and $B(x)$. To be more powerful for free, we can use the paired variance

$$\mathrm{Var}_x[A(x) - B(x)] = \mathrm{Var}_x[A] + \mathrm{Var}_x[B] - 2\mathrm{Cov}_x[A(x), B(x)].$$

The Cov term may potentially reduce the paired variance to 0 when A and B are perfectly correlated.

From the standard errors we can obtain the z-score $z = \frac{\bar{A}-\bar{B}}{\mathrm{SE}[A-B]}$. When $N > 100$ is moderately large, then the Central Limit Theorem applies and the z-scores can be converted to p-values. For example,

$\Pr[|z| > 1] = 0.32$ is very weak, $\Pr[|z| > 5] < 10^{-6}$ is beyond doubt, and $\Pr[|z| > 1.96] = 0.05$ for a p-value of 0.05 is a reasonable conventional standard. Once these standards are intuitive, we can just work with the better behaved z-scores directly.

3.2 Sampling multiple predictions on each question

To capture this setting, we use ϵ for the seed generating the noise, which is independent of the question x, thus allowing the concise notation $\mathrm{Var}_x[\mathrm{E}_\epsilon[A]] = \mathrm{Var}_x[E_{\epsilon|x}[A(x,\epsilon) \mid x]]$. As before, but now also averaging over the K samples for each question x, we consider the average score

$$\bar{A} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{K}\sum_{j=1}^{K} A(x_i,\epsilon_{ij}) \approx \mathrm{E}_{x,\epsilon}[A(x,\epsilon)],$$

The estimated variance is then

$$\mathrm{Var}_{x,\epsilon}[A(x,\epsilon)] \approx \frac{1}{N}\sum_{i=1}^{N}\frac{1}{K}\sum_{j=1}^{K}(A(x_i,\epsilon_{ij}) - \bar{A})^2.$$

We have the law of total variance satisfied by the components

$$\mathrm{Var}_{x,\epsilon}[A] = \mathrm{Var}_x[\mathrm{E}_\epsilon[A]] + \mathrm{E}_x[\mathrm{Var}_\epsilon[A]].$$

$\mathrm{Var}_{x,\epsilon}[A]$ is the total variance of first drawing a question x from the pool of questions, and then drawing a sample answer for this question. $\mathrm{Var}_x[\mathrm{E}_\epsilon[A]]$ is variance of the expected score $\mathrm{E}_\epsilon[A]$, which we call the data noise. $\mathrm{E}_x[\mathrm{Var}_\epsilon[A]]$ is the variance due to sampling from the model, which we call the prediction noise. See A.1 for a discussion on why $\mathrm{E}_x[\mathrm{Var}_\epsilon[A]]$ is the correct prediction variance and its other decomposition $\mathrm{Var}_\epsilon[\mathrm{E}_x[A]]$.

Thus all noises can be normalized to be the standard errors,

$$\begin{aligned}
\mathrm{SE}(A,N) &= N^{-1/2}(\mathrm{Var}_{x,\epsilon}[A])^{1/2},\\
\mathrm{SE}_x(A,N) &= N^{-1/2}(\mathrm{Var}_x[\mathrm{E}_\epsilon[A]])^{1/2},\\
\mathrm{SE}_{\text{pred}}(A,N) &= N^{-1/2}(\mathrm{E}_x[\mathrm{Var}_\epsilon[A]])^{1/2}.
\end{aligned}$$

Paired comparison. When we have two models A and B, we can consider the paired standard deviation $\mathrm{Var}_{x,\epsilon}[A(x,\epsilon) - B(x,\epsilon)]$, which also satisfy the law of total variance on $A - B$,

$$\mathrm{Var}_{x,\epsilon}[A - B] = \mathrm{Var}_x[\mathrm{E}_\epsilon[A - B]] + \mathrm{E}_x[\mathrm{Var}_\epsilon[A - B]]. \tag{1}$$

3.3 Estimating from samples

While (1) allows us to estimate the variance components directly from K paired samples $A(x,\epsilon_j) - B(x,\epsilon_j)$ for $j = 1,\ldots,K$, this is inaccurate since the seeds or predictions are interchangeable. We can use this to derive formulas for estimating from samples accurately, as if using all $K \times K'$ pairs of collected predictions $A(x,\epsilon_j) - B(x,\epsilon'_k)$ for $j = 1,\ldots,K$ and $k \in 1,\ldots,K'$. First, the prediction variance $\mathrm{E}_x[\mathrm{Var}_\epsilon[A - B]]$ decomposes by independence on each question x,

$$\begin{aligned}
\mathrm{Var}_\epsilon[A(x,\epsilon) - B(x,\epsilon')] &= \mathrm{Var}_\epsilon[A(x,\epsilon)] + \mathrm{Var}_\epsilon[B(x,\epsilon)],\\
\mathrm{E}_x[\mathrm{Var}_\epsilon[A(x,\epsilon) - B(x,\epsilon')]] &= \mathrm{E}_x[\mathrm{Var}_\epsilon[A(x,\epsilon)]] + \mathrm{E}_x[\mathrm{Var}_\epsilon[B(x,\epsilon)]].
\end{aligned}$$

Next, the data variance $\mathrm{Var}_x[\mathrm{E}_\epsilon[A - B]] = \mathrm{Var}_x[\mathrm{E}_\epsilon[A] - \mathrm{E}_\epsilon[B]]$ decomposes by linearity of expectation. Finally, the total variance is slightly tricky, where

$$\begin{aligned}
\mathrm{Var}_{x,\epsilon}[A - B] &= \mathrm{Var}_{x,\epsilon}[A] + \mathrm{Var}_{x,\epsilon}[B] - 2\mathrm{Cov}_{x,\epsilon}[A,B]\\
&= \mathrm{Var}_{x,\epsilon}[A] + \mathrm{Var}_{x,\epsilon}[B] - 2\mathrm{Cov}_x[\mathrm{E}_\epsilon[A],\mathrm{E}_\epsilon[B]].
\end{aligned} \tag{2}$$

The equality is by the law of total covariance, $\mathrm{Cov}_{x,\epsilon}[A,B] = \mathrm{Cov}_x[\mathrm{E}_\epsilon[A],\mathrm{E}_\epsilon[B]] + \mathrm{E}_x[\mathrm{Cov}_\epsilon[A,B]]$ where the second term is 0 since different random predictions on a given x are independent.

Small K correction. To estimate $\mathrm{Var}_x[\mathrm{E}_\epsilon[A]]$ from K sampled predictions per question, a correction from the direct estimator can significantly increase the accuracy. We describe the concept for the unpaired case for simplicity. By the law of total variance on $\bar{A}_K(x,\epsilon) = \frac{1}{K}\sum_{j=1}^{K} A(x,\epsilon_j)$, we have

$$\begin{aligned}
\mathrm{Var}_{x,\epsilon}[\bar{A}_K] &= \mathrm{Var}_x[\mathrm{E}_\epsilon[\bar{A}_K] + \mathrm{E}_x[\mathrm{Var}_\epsilon[\bar{A}_K]] \\
&= \mathrm{Var}_x[\mathrm{E}_\epsilon[A]] + \frac{1}{K}\mathrm{E}_x[\mathrm{Var}_\epsilon[A]] \\
\implies \mathrm{Var}_x[\mathrm{E}_\epsilon[A]] &= \mathrm{Var}_x[\bar{A}_K] - \frac{1}{K}\mathrm{E}_x[\mathrm{Var}_\epsilon[A]] \\
&\approx \hat{\mathrm{Var}}_x[\bar{A}_K] - \frac{1}{K-1}\hat{\mathrm{E}}_x \hat{\mathrm{Var}}_\epsilon[\bar{A}_K]
\end{aligned}$$

where the hats mean sample estimate. This correction for small K can be important because we will average over N questions. If the estimate on each question is biased by $1/(K-1)$, the average error will still be off by the bias no matter how big N is. Whereas if the estimate is unbiased on each question, then the average error can decrease as N increases. Figure 3 shows this observation on MATH500, where the small-K correction is needed to reduce the error to an acceptable level even with 2000 bootstrapped questions. The high relative error is because the paired data noise is much smaller than the prediction noise. An instructive example is to consider when $\mathrm{E}_\epsilon[A-B]=0$ but an estimate using K samples will not be 0 due to non-zero prediction noise.

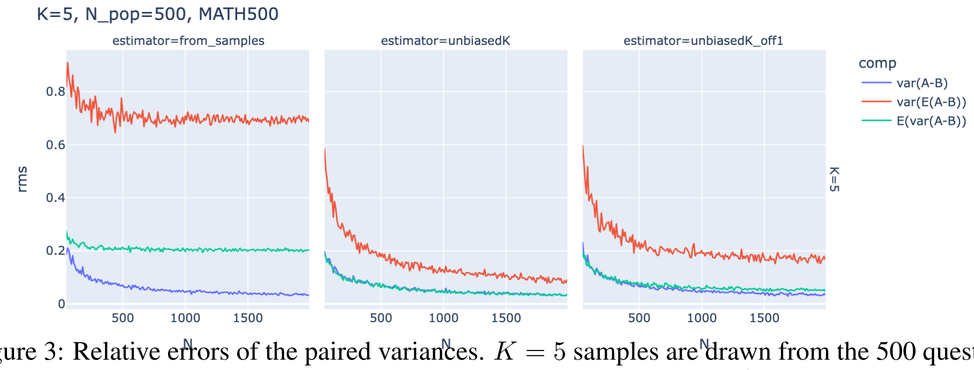

Figure 3: Relative errors of the paired variances. $K=5$ samples are drawn from the 500 questions from MATH500, where 1000 real predictions are drawn from models A and B on each question, which is treated as the ground truth. The root mean squared relative errors (rms) of variance components are plotted vs. bootstrap sample size N, extending beyond the population size 500. The left figure without the correction on K has an unacceptably high 70% relative error even with 2000 data points. The middle figure is with the correction of §3.3 and the right figure uses $1/K$ instead of $1/(K-1)$.

3.4 Implementation in array notation

In an experiment, we typically evaluate the model on all N questions, and may draw K answers for each question. In this section, we overload our notation with actual samples $A, B \in \mathbb{R}^{N\times K}$ and present the formulas in Section 3.3 in numpy-style code. For simplicity, we use the same number of predictions $K_i = K$ on all questions x_i. For the unpaired case, we have Table 1. For paired, we have Table 3.

3.5 Equivalence to the bootstrap and the sign test

Two other very well-established methods are the bootstrap (Efron and Tibshirani, 1986) and paired difference tests such as the sign test (Dixon and Mood, 1946). These methods ask if the observed results are likely when the examples are random for bootstrap, and when the comparison outcomes are random for the sign test. In this section we show they all give the same answer.

Bootstrap is a general method where resampling the given examples uniformly with replacement is shown to give the right answer for many estimation problems, including estimating the standard error. That is, resampling the given samples uniformly with replacement gives the same answer as getting more real samples from the population. Applied to the problem of noise, the natural question is how likely we are to get the opposite result $\mathrm{E}[A] < \mathrm{E}[B]$ as opposed to the observed

Name	Formula	Code	Bernoulli
total variance	$\mathrm{Var}_{x,\epsilon}[A]$	`var(A)`	$\bar{p}(1-\bar{p})$
data variance	$\mathrm{Var}_x[\mathrm{E}_\epsilon[A]]$	`var(mean(A, axis=1))-b`	$\frac{1}{N}\sum_i (p_i - \bar{p})^2$
prediction variance	$\mathrm{E}_x[\mathrm{Var}_\epsilon[A]]$	`mean(var(A, axis=1))+b`	$\frac{1}{N}\sum_i p_i(1-p_i)$

where `b=1/(K-1)*mean(var(A, axis=1))`

Table 1: Unpaired estimators where $A \in \mathbb{R}^{N\times K}$ with K samples for each of N questions.

Formula	Code
$\mathrm{Var}_{x,\epsilon}[A-B]$	`var(A)+var(B)-2*cov(mean(A,axis=1),mean(B,axis=1))`
$\mathrm{Var}_x[\mathrm{E}_\epsilon[A-B]]$	`var(mean(A,axis=1)-mean(B,axis=1))-b`
$\mathrm{E}_x[\mathrm{Var}_\epsilon[A-B]]$	`mean(var(A,axis=1)+var(B,axis=1)) +b`

where `b=1/(kA-1)*mean(var(A, axis=1))+1/(kB-1)*mean(var(B, axis=1))`

Table 2: paired estimators where $A \in \mathbb{R}^{N\times K_A}$, $B \in \mathbb{R}^{N\times K_B}$.

result $\mathrm{E}[A] > \mathrm{E}[B]$. If the opposite outcome also has significant probability, then the conclusion is not statistically significant.

The sign test predates bootstrap and instead of sampling the questions, it makes the comparison outcome random. It supposes that any difference actually happens with a random probability $\Pr[A(x) \neq B(x)] = 0.5$ (null hypothesis) and asks how likely we are to get a more extreme outcome than what we actually observe. If it is quite likely to observe a more extreme outcome under the null hypothesis, then the conclusion is not statistically significant.

The key step in both methods is estimating the variance of their respective random procedure. If the variances are the same, then the probability to observe the opposite result from an observed mean (bootstrap) is also the same as the probability of observing a more extreme result if the real mean is 0 (sign test). The details are deferred to B.

4 Experiments

The experimental results of this paper can be found at `https://all-the-noises.github.io`, where the method is tested on more benchmarks and where the figures are interactive. We can use this to check that the main conclusion indeed holds on all the evals.

4.1 Exploratory analysis and findings

4.1.1 Data heatmap

To get an overview of our data, we use a heatmap shown in Figure 4. Each row is a different example, sorted from the easiest to the hardest. Each column is a model whose x-coordinate is the overall accuracy $\mathrm{E}[A]$ of this model. Like models, we might consider showing examples with y-coordinate at their average accuracy instead of rank, but that would depend on which models are included, and less intrinsic to the benchmark. We immediately observe that models at a similar overall accuracy also does similarly on individual questions. Around 1000 samples are drawn for each question and model for MATH500 in Figure 4. The SWEbench figure is based on the leaderboard, which had to commit to single answers so it is binary, but the same overall pattern holds, though with more noise. On MATH500, most of the red regions of bad models are not 0, but are rather a few percent accuracy.

4.1.2 All-pairs paired noises and predictable total noise

Figure 1 shows that the total SE agrees well with the $\mathrm{Beta}(p, 1-p)$ theory predictions described in 4.1.3 and the range of the data noise and prediction noises. Only models pairs that are close in performance are plotted $E[A] - E[B] < 5\,\mathrm{SE}(A-B)$ to avoid pointless comparisons between models whose difference is not in doubt.

The noise components can be traded off so they can depend on the setting and not as predictable. We find at more natural temperatures of 0.6-1, the prediction noise tend to be higher than the data noise. Figure 5 shows that the prediction noise dominates at 0.8 temperature, whereas the data noise dominates at 0.2 temperature, and yields about the same total noise.

4.1.3 The Beta theory makes good noise predictions and fits the data

The Beta theoretical model says that the expected accuracy of each question follows the $\mathrm{Beta}(p, 1-p)$ distribution where p is the mean of the model. More precisely, the expected accuracy of question x_i

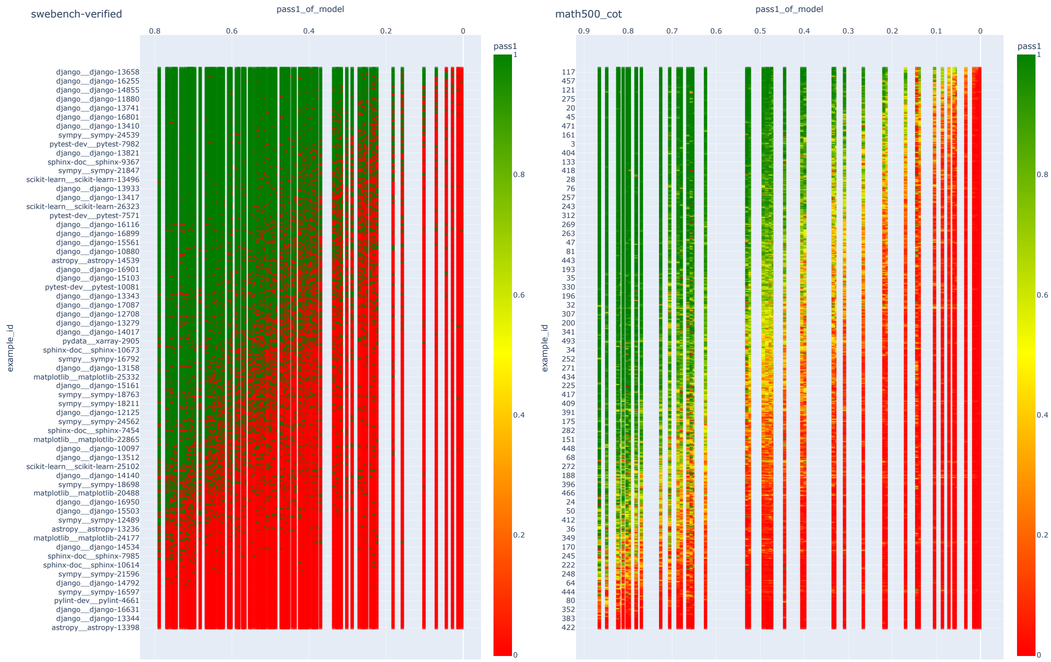

Figure 4: Results heatmap. Each row is a different example, sorted from the easiest to the hardest. Each column is a model whose x-coordinate is the overall accuracy. Left: SWEbench-verified, 1 prediction per question; Right: MATH500, 1000 predictions per question. See Section 4.1 for more details.

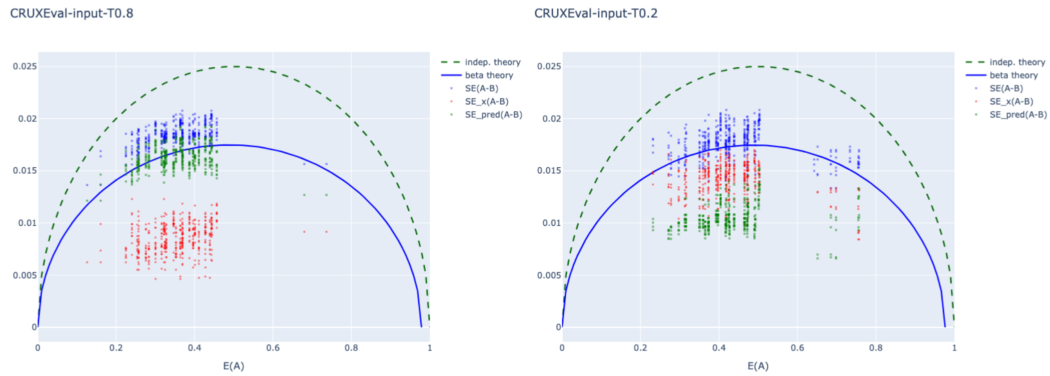

Figure 5: All noise components on CRUXEval at temperature 0.8 (left) and 0.2 (right). The prediction tend to be higher at natural temperatures 0.7-1.

is $u_i \sim \text{Beta}(p, 1-p)$ so that $A(x_i), B(x_i) \sim \text{Bernoulli}(u_i)$ and $\text{E}[A(x_i)] = u_i$. Too see that this leads to the observed SE, the models draw independent predictions on each x_i so $\text{Cov}_\epsilon[A, B \mid x_i] = 0$. Thus we have $\text{E}_x[\text{Var}_\epsilon[A - B]] = \text{E}_x[2\text{Var}_\epsilon[A]]$, which integrates to

$$\int_0^1 2u(1-u) \frac{u^{p-1}(1-u)^{(1-p)-1}}{B(p, 1-p)} du = \frac{B(p+1, (1-p)+1)}{B(p, (1-p))} = p(1-p),$$

where $B(a, b) = \int_0^1 u^{a-1}(1-u)^{b-1} du$ is the normalization constant for the $\text{Beta}(a, b)$ distribution. One consideration to fit the data better is that all models fail on some questions, so we suppose those examples actually have $u_i = 0$ instead of $\text{Beta}(p, 1-p)$. Figure 6 shows the quality of fit using $\text{Beta}(a, b)$. Smaller $a + b$ means more bimodal where $a + b$ tend to decrease when the model accuracy is higher, showing that better models become more bimodal whereas very bad models is usually unimodal. For an extreme example, guessing uniformly on multiple choice questions with C choices means the expected accuracy $u_i = 1/C$ for all questions, which is has a single mode at $1/C$.

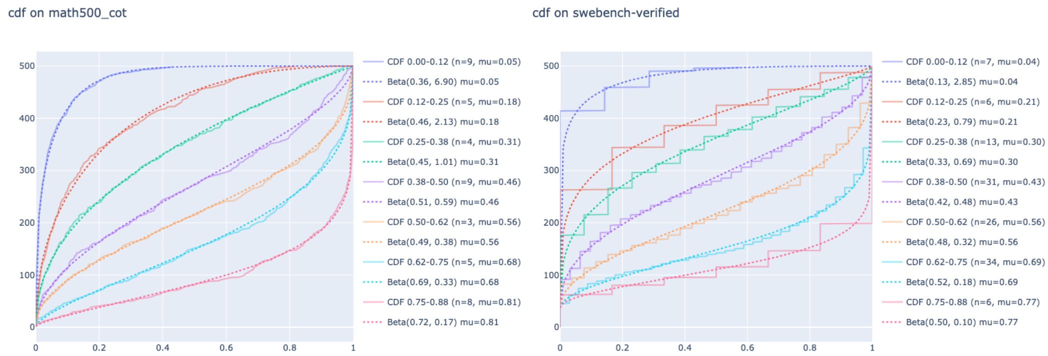

Figure 6: Empirical cummulative distribution curves (CDF, solid) and their respective Beta models (dots). Each color represents a different bin by model accuracy. The average accuracy of each question in each bin of models is computed and sorted to produce the CDF curves. The fit is good.

5 Related work

Much work deal with noise in science and machine learning (Lehmann and Romano, 2005; Dror et al., 2018; Hermann et al., 2024; Card et al., 2020, among many others), we adopt the approach of Miller (2024) to estimate the variance directly. While using samples to estimate the population follows from the law of large numbers, their approach is clarifying, has unique considerations for LLMs, and is directly generalizable all metric functions. Their method is also consistent with the bootstrap (Efron, 1979) and the sign test (Dixon and Mood, 1946) when applicable (Section 3.5). Efron and Tibshirani (1986) clearly described estimating the standard error as the example of not needing bootstrap. The concept of total variance decomposition is present the *analysis of variance*, but Section 3 of Miller (2024) applied this concept to LLMs under *variance of the conditional mean* and *mean conditional variance*, which we call the prediction noise and the data noise. In this work, we develop and test the methods of estimating noise types from samples and then empirically measure these variance components on all pairs of models and draw some general conclusions from the observations.

For leaderboards, Chatbot arena (Chiang et al., 2024) and CRUXEval (Gu et al., 2024) computed confidence intervals using bootstrap with a fixed reference model. Their approach is needed for per-model confidence intervals but is incorrect (Example 4). Furthermore, Chatbot arena aggregates comparisons with all other models to compute their confidence intervals, which necessarily loses any special properties of a particular pair of models. This work actually compares all pairs of models, and summarize all the comparisons to show that the total noise level is predictable on LLM benchmarks.

Madaan et al. (2024) measured the noise due to random seeds. Any direct approach to measure the noise necessarily ignores the data noise (Example 1). For the data noise, they also proposed using unpaired Bernoulli confidence intervals. This method was adopted by Llama 3 (Grattafiori et al., 2024) but is much looser than the paired methods (Example 2) and does not separate the data noise and the prediction noise. Improving model ability is hard and expensive while tightening the analysis is well-established. This work shows it can be done for all results together and decoupled from the main experiments.

6 Discussions

6.1 Why is noise predictable

The range of correctness evaluations $A(x) \in [0, 1]$ immediately give an meaningful upperbound on the noise $\mathrm{SE}[A] \le (\frac{1}{N} 1/4)^{1/2}$ and $\mathrm{SE}[A - B] \le (\frac{2}{N} 1/4)^{1/2}$. The dependence on the mean accuracy p_A is $\mathrm{SE}[A - B] \le (\frac{2}{N} p_A(1 - p_A))^{1/2}$, which is shown by the independent theory curve of Figure 1. We should expect LLM predictions to be positively correlated because questions have different intrinsic difficulties and LLMs are also trained on similar data, thus we expect $\mathrm{SE}[A - B]$ to be smaller than the independent curve.

The lowerbound is $\mathrm{SE}[A - B] = 0$ if $A(x) = B(x)$ on all x. For a strict net improvement on k questions bounded by a constant multiple of the standard error, $\mathrm{SE}[A - B] = O(N^{-3/4})$, still much smaller than the actual trend of $\mathrm{SE}[A - B] = O(N^{-1/2})$. In Figure 1, swebench-verified, we have 2 results near $x = 0.3$ that comes from SWE-Fixer (Xie, 2025), due to using a deterministic filter. Another explanation is that the prediction noise alone is also $O(N^{-1/2})$. The Beta model provides a partial explanation too, where all the between close pairs are hypothesized to come from prediction. Still, on Figure 4, it is clear there is some data noise even between close pairs. On leaderboard predictions such as SWEBench (Jimenez et al., 2024), or LiveCodeBench (Jain et al., 2024), there usually is only $K = 1$.

6.2 Solving hard and special problems

For a problem requiring a long answer where guessing correctly is unlikely, answering even 1 problem might be significant and interesting. For example, the problem can ask for the proof of an important open problem and the test checks the proof. If a model (or someone) solves such a hard problem then we should not object to the sample size of 1. We probably don't need to consider this yet. In the empirical data, all pairs of models have enough noisy inconsistencies where model A may beat B on 2 hard examples, but then B beats A on 2 easy examples. If A is so good that it didn't make any mistakes on the long complex answer required to solve the hard problems, why did it fail on some easy ones? Second, problems solvable by mediocore models or where reference solutions can be found on the internet (i.e. training data) is unlikely to deserve special deference, which still applies to most evaluation sets.

In our human experience, we intuitively get a lot of signals by evaluating on hard problems, for example in interviews. The key is perhaps gaining much more information than just one bit of binary correctness judgement from the details of how the problem was solved. In contrast, LLM evaluations and learning settings only give back 1 bit after a lot of work. If we move in the direction of smaller and harder evals, we probably need to also generate more information per question.

6.3 Caveats of the expected metric.

While we can recommend the expected metric for controlled experiments testing a single factor, there are some caveats when trying to improve an eval using the best available methods. Suppose we have 100 True/False questions, model A has an expected accuracy of 60% and Model B 61% on each question.

Expected metric. We draw enough samples per question to be sure that A is 60% and that B is 61%. **Conclusion**: A and B are significantly different. This metric is used by many papers of the area, and corresponds to the *data noise* after averaging away some prediction noise.

Best prediction metric. Since the probability of correct is greater than half, both models can achieve 100% accuracy if they predict by majority voting or thresholding. **Conclusion**: A is the same as B. This is used by most classification evaluations and should be used when pushing the benchmark.

One prediction metric. One sampled prediction per question is evaluated, so A and B do about 60% with a 7% standard error on the difference. **Conclusion**: A is not significantly different from B. This metric is based on a single answer per question, is used in most of the sciences and tests for humans, and corresponds to the *total noise*.

If our goal is do push the state-of-the-art of a particular eval using whatever method, examples 5 and 6 show how the expected accuracy can be misleading and the best prediction should be obtained first as the baseline, otherwise sharpening the distribution or including better in-domain data can all be expected to produce a small but significant result. Example 7 shows that arbitrarily small but consistent changes is significant with expected prediction, clarifying that a standard of effect size is also needed, for the real world may not be impressed by a 1% expected improvement even if it is definitely *not* noise.

Example 5 (majority voting)**.** Suppose the expected accuracies on all questions are the same with $\mathrm{E}[A] = 0.6$, $\mathrm{E}[B] = 0.9$ where the expectation averages out the prediction noise. This big improvement in expected accuracy might be trivially achievable by majority voting if we try to put forward the best prediction instead. Whenever the equivalence of sampled predictions can be checked, majority voting can increase the accuracy to 100% if the equivalence class with the correct answer has the highest probability, which is the case when $\mathrm{E}[A] > 0.5$.

Example 6 (reliable verification)**.** Suppose that on half of the questions $\mathrm{E}[A] = 0.1$ and $\mathrm{E}[B] = 1$, but on the other half of questions $\mathrm{E}[A] = 0.1$ and $\mathrm{E}[B] = 0$. While $\mathrm{E}[B] = 0.5 > \mathrm{E}[A] = 0.1$ with B having a much higher average, A is better if the predictions can be verified (search problems or formal proofs) where A is 100% with a verifier but B is only 50%. If a reliable verifier is available, then the *pass-at-large-K* metric is a more meaningful metric than the average.

Example 7 (Arbitrarily small but consistent improvement is significant.)**.** Suppose that on question x_i, $\mathrm{E}[A(x_i)] = p_i$, $\mathrm{E}[B(x_i)] = p_i + e$, then with enough samples we can conclude that A and B are significantly different no matter how small e is.

6.4 Applications and Recommendations

We advocate for more specialization and the decoupling of noise analysis from experiments, so practitioners can focus on their work instead of doing correct noise analysis. The noise analysis more naturally fall on benchmark and leaderboard builders. Leaderboard and benchmark builders can run the all-pairs-paired method to measure all the noise types on their benchmark, thus taking more responsibility for the reliability of their benchmark. This can even be done by a third-party as in this work, as long as leaderboard maintainers release their question level results like Jimenez et al. (2024); Jain et al. (2024) ideally with multiple predictions per question so the noise components can be better estimated.

If we need to squeeze out more of the evaluations, practitioners can also use our method on their own set of results to get more sensitivity while ensuring small difference are still unlikely to be noise.

Noise-aware Meta-analysis of multiple evals. We have a lot evaluations of varying quality and size, and many huge table of results that are hard to interpret. Averaging MMLU where $N \sim 10k$ and HumanEval where $N \sim 100$ likely leads to worse results than using MMLU alone. Instead of the absolute improvement, the percent relative improvements, or the confidence intervals, we advocate using the z-score because it is signed and more intuitive, and is well-behaved. Recall the z-score $z = \frac{\bar{A}-\bar{B}}{\mathrm{SE}[A-B]}$. This work measured the paired and unpaired versions of prediction, data, and total SEs and is suitable here. Inspecting this list of z-scores (one for each benchmark) and checking their consistency is already informative. To get a meta-z-score given the z-score of benchmark i, an intuitive method is $\bar{z} = \frac{\sum_i w_i z_i}{(\sum_i w_i^2)^{1/2}}$, where some prior beliefs can be incorporated into w_i. For example, $w_i = 1$ means that we believe each benchmark has equal weight, whereas $w_i = N_i^{1/2}$ means each question has equal weight. Probably incorporating our prior belief of the importance of each benchmark into the weight is reasonable here while staying around 1. See Zaykin (2011) for more related work and analysis, on which we have two remarks: the scaling law can be a source of the expected effect size on a benchmark if desired; giving each question from different evals the same weight is less valid even if it has the most statistical power under their assumptions.

6.5 Limitations

While the method is general, all the empirical data are from correctness evals, so whether the empirical finding generalize to other evals remains to be determined. Bowyer et al. (2025) show that CLT does not work well for less than a few hundred examples, though with 100 questions the errors are only significant in the tails. For simplicity, we did not consider the potentially impactful clustered correction of (Miller, 2024) though incorporating it does not pose conceptual difficulty.

Acknowledgments. I thank Sean O'Brien, Lovish Madaan, Dieuwke Hupkes, Alex Gu, Jiawei Liu, Yuhang Lai, and Sten Sootla for making question-level data available for analysis. I am grateful to Evan Miller, Nicolas Usunier, Zach Rait, Yuxiang Wei, Jannik Kossen, and Ari Holtzman for valuable discussions and feedback; Pedro Rodriguez, Ofir Press, Naman Jain, Baptiste Rozière, and Gabriel Synnaeve for their advice and support. The all-pairs approach is inspired by Chatbot Arena.

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A Analysis details

A.1 Alternative decomposition

Taking expectation over either the noise or the question has the same total variance and thus the sums are equal

$$\begin{aligned}\mathrm{Var}_{x,\epsilon}[A] &= \mathrm{Var}_x[\mathrm{E}_\epsilon[A]] + \mathrm{E}_x[\mathrm{Var}_\epsilon[A]] \\ &= \mathrm{Var}_\epsilon[\mathrm{E}_x[A]] + \mathrm{E}_\epsilon[\mathrm{Var}_x[A]]\end{aligned}$$

To see that it indeed corresponds to what we want to measure,

$$\mathrm{Var}_\epsilon\left[\frac{1}{N}\sum_{i=1}^{N} A(x_i, \epsilon_i)\right] = \frac{1}{N^2}\sum_{i=1}^{N} \mathrm{Var}_\epsilon[A(x_i, \epsilon)] \to \frac{1}{N}\mathrm{E}_x[\mathrm{Var}_\epsilon[A(x, \epsilon)]. \tag{3}$$

The left hand side is the prediction noise on N particular data points, which approach the mean if they are iid. One can directly evaluate the prediction noise on a particular dataset by running the eval K times and measure the variance of the K results. That is also correct in expectation but could be much more noisy since (3) includes equally valid samples not drawn by any particular evaluation run.

Paired prediction noise from the same model Using (2) to compute $\mathrm{Var}_{x,\epsilon}[A(x, \epsilon_1) - A(x, \epsilon_2)]$ is too small when using the same set of samples for A. Instead, we can use the following equality and an unbiased estimator for $\mathrm{E}_x[\mathrm{Var}_\epsilon[A]]$ with $A = A(x, \epsilon_1)$, $B = A(x, \epsilon_2)$.

$$\begin{aligned}\mathrm{Var}_{x,\epsilon}[A - B] &= \mathrm{Var}_x[\mathrm{E}_\epsilon[A - B]] + \mathrm{E}_x[\mathrm{Var}_\epsilon[A - B]] \\ &= 0 + \mathrm{E}_x[\mathrm{Var}_\epsilon[A] + \mathrm{Var}_\epsilon[B]]] \\ &= 2\mathrm{E}_x[\mathrm{Var}_\epsilon[A]]\end{aligned}$$

Equivalently, we can still use (2) and Table 3 if we sample without replacement when estimating $\mathrm{E}[A(x, \epsilon_1)A(x, \epsilon_2)]$ to avoid the bias over-estimating the covariance using the same set of samples. Suppose we have K samples for a question x with scores $A_1, \ldots, A_k$. To estimate $\mathrm{E}[A(x, \epsilon_1)A(x, \epsilon_2)]$ we must use $\texttt{mean}_{i,j\neq i} A_i A_j = \frac{1}{K^2-K}\sum_{i,j\neq i} A_i A_j = \frac{1}{K^2-K}[(\sum_i A_i)^2 - \sum_i A_i^2]$.

B Equivalence to bootstrap and sign test

To setup both methods, let $W_A := \sum_{i=1}^{N} \mathbb{I}[A(x_i) > B(x_i)]$ be the number of times model A wins against model B and vice versa for W_B, and let's assume $W_A > W_B$ for convenience. The question is how likely we have the same conclusion $W_A > W_B$ under the randomness prescribed by bootstrap or the sign-test.

The bootstrap. We draw another N samples with replacement from the existing samples $x_1, \ldots, x_N$ to obtain $X_A, X_B, X_0 \sim \mathrm{multinomial}(N, q_A, q_B, q_0)$ with $X_A + X_B + X_0 = N$, where the outcome probabilities are $q_A = W_A/N$ for A win, $q_B = W_B/N$ for B win, and $q_0 = 1 - q_A - q_B$ for tie. The mean and variance of the resamples are respectively $E[X_A - X_B] = W_A - W_B > 0$,

$$\begin{aligned}\mathrm{Var}[X_A - X_B] &= \mathrm{Var}[X_A] + \mathrm{Var}[X_B] - 2\mathrm{Cov}[X_A, X_B] \\ &= N\left(q_A(1 - q_A) + q_B(1 - q_B) + 2q_A q_B\right) \\ &= N\left(q_A + q_B - (q_A - q_B)^2\right) \\ &\approx N(q_A + q_B) = W_A + W_B.\end{aligned} \tag{4}$$

Under bootstrap, a natural question is how likely we would see the opposite result $\Pr[X_A < X_B]$. With an approximation that $(q_A - q_B)^2 \ll q_A + q_B$ (else the result is probably significant already), the bootstrap asks if $\Pr[z \geq \frac{W_A - W_B}{\sqrt{W_A + W_B}}]$ for standard normal z.

Directly estimating the variance from the samples also gives the same answer as (4). While this is a consequence of bootstrap variance equal to the sample variance, a direct calculation is this

$$\begin{aligned}\mathrm{Var}_x[A(x) - B(x)] &= E[(A - B)^2] - E[A - B]^2 \\ &= q_A + q_B - (q_A - q_B)^2.\end{aligned}$$

The sign test. When comparing model A vs. model B, the null-hypothesis is that model A and B each has a 1/2 chance of being better (win) on each example. A wins if A gets a question correct but B does not, tie if A and B are both correct or incorrect. The question is if the observed results are likely to happen under this null-hypothesis.

The p-values is then $\Pr[X \geq W_A]$ for $X \sim \text{binomial}(W_A + W_B, 0.5)$ with $E[X] = \frac{1}{2}(W_A + W_B)$ and $\text{Var}[X] = \frac{1}{4}(W_A + W_B)$. The question is how likely is X to be as extreme as W_A as observed. When $W_A + W_B$ is large, the normal approximation is accurate and reduces to $\Pr[z > \frac{W_A - W_B}{\sqrt{W_A + W_B}}]$ for standard normal z (i.e. one sided). If we also consider how A might be worse, then we should consider the two-sided question.

So the main empirical variance method, the bootstrap, and the sign test give the same answer. A slight approximation was used on the sign test. Without the approximation, $\text{Var[sign test]} \geq \text{Var[bootstrap]}$ due to using 0.5 for the null hypothesis instead of the empirical win rate.

Formula	**Win rates**
$\text{Var}_{x,\epsilon}[A - B]$	$q_A + q_B - (q_A - q_B)^2$
$\text{Var}_x[\text{E}_\epsilon[A - B]]$	$\frac{1}{N}\sum_i q_{A,i} + q_{B,i} - (q_A - q_B)^2$
$\text{E}_x[\text{Var}_\epsilon[A - B]]$	$\frac{1}{N}\sum_i q_{A,i} + q_{B,i} - (q_{A,i} - q_{B,i})^2$

Table 3: For where $A, B \in \{0, 1\}$ with $\Pr[A(x_i) > B(x_i)] = q_{A,i}$.

C Estimator testing

We validate our variance estimators by sampling from generative models where the ground truth is known. Our primary focus is measuring estimation accuracy and how it scales with sample size, though we also check the bias and test some properties as well.

For each test scenario, we generate data $A, B \in \mathbb{R}^{N \times K}$ from models with known variance components, apply our estimators, and compare against ground truth over many independent trials. We use two primary metrics: (1) root mean square (rms) relative error to measure accuracy, and (2) t-scores to detect systematic bias.

We test under three generative processes. The *Bernoulli model* samples N problems with replacement from a population parameterized by $p_A \in \mathbb{R}^{N_{\text{population}}}$, generating K Bernoulli trials per problem. The *stratified Bernoulli model* evaluates when every question is used exactly once, simulating the case of drawing samples from real evaluation, breaking independence and introduces tricky effects. The *bootstrap model* resamples from empirical metric matrices, useful for comparisons to bootstrap and testing based on real data.

Accuracy vs. sample size. Our main concern is practical accuracy. Tests establish that estimators achieve rms error below 0.25 for $N = 100$ sample sizes and below 0.13 for larger samples ($N = 400$). We verify this across diverse probability distributions including uniform, Beta, specific distributions and real data. For paired comparisons, tests confirm we get reasonable accuracy (relative rms < 0.1) on the total variance and prediction variance. The data variance is the hardest to estimate since it tend to be the smallest where correct for the bias in K is found to be critical by such testing (Figure 3).

Bias Testing. Deriving and testing unbiased estimators proved useful for understanding estimator for each type of sampling procedure. The process of eliminating bias forced us to carefully account for all sources of estimation error—finite sample corrections in both N and K, paired sampling effects, and stratified sampling adjustments. Moderately large N is necessary for accurate estimation, but we test corrections for N as well. On the other hand, small K is common in practice, so we focus on K, and found it to be more than just a bit of bias correction but critical in the case of large N but small K.

This testing builds confidence that our estimators behave as predicted and that we understands how to get unbiased estimators in each situation, even as we only recommend two of the simplest estimators.